

STEP 1: Create Project Folder

```
mkdir Program-8  
cd Program-8
```

STEP 2: Create Node.js App (server . js)

```
nano server.js
```

```
cat: server.js: No such file or directory  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program8$ cat server.js  
const http = require("http");  
const port = 3000;  
  
const server = http.createServer((req, res) => {  
  res.end("Hello program8");  
});  
  
server.listen(port, () => {  
  console.log(`Server running at port ${port}`);  
});  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program8$
```

STEP 3: Create ConfigMap

```
1RV24MC087_SSDhanush@dhanush-Latitude:~/program8$ cat nodejs-configmap.yaml  
apiVersion: v1  
kind: ConfigMap  
metadata:  
  name: nodejs-app-config  
data:  
  server.js: |  
    const http = require("http");  
    const port = 3000;  
  
    const server = http.createServer((req, res) => {  
      res.end("Hello program8");  
    });  
  
    server.listen(port, () => {  
      console.log(`Server running at port ${port}`);  
    });  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program8$ cat nodejs-end.yaml
```

STEP 4: Create Pod YAML

```
});  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program8$ cat nodejs-pod.yaml  
apiVersion: v1  
kind: Pod  
metadata:  
  name: program-8  
  labels:  
    app: nodejs  
spec:  
  containers:  
    - name: nodejs  
      image: node:18  
      command: ["node", "/usr/src/app/server.js"]  
      volumeMounts:  
        - name: app-code  
          mountPath: /usr/src/app  
      ports:  
        - containerPort: 3000  
  volumes:  
    - name: app-code  
      configMap:  
        name: nodejs-app-config  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program8$ cat server.yaml
```

STEP 5: Apply YAML Files

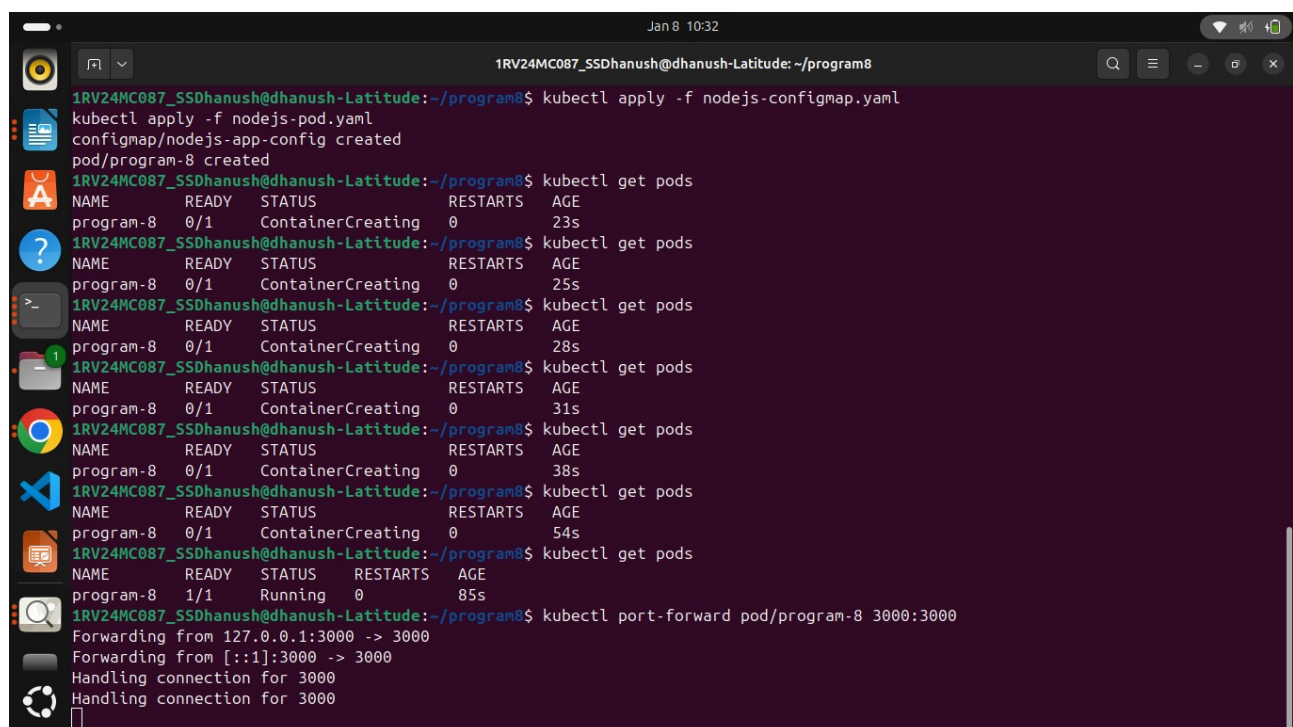
```
kubectl apply -f nodejs-configmap.yaml  
kubectl apply -f nodejs-pod.yaml
```

STEP 6: Check Pod Status

```
kubectl get pods
```

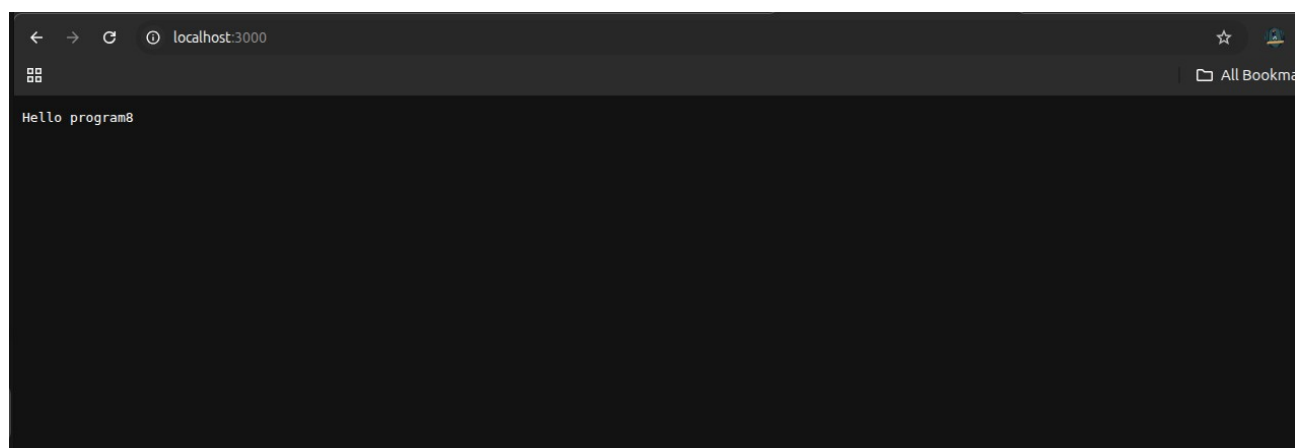
STEP 7: Access the Node.js App

```
kubectl port-forward pod/program-8 3000:3000
```



A terminal window titled "1RV24MC087_SSDhanush@dhanush-Latitude: ~/program8" showing the execution of kubectl commands. The output shows the creation of a configmap and a pod, followed by multiple checks of the pod status. The pod "program-8" is initially in "ContainerCreating" status and eventually reaches "Running" status. The final command is "kubectl port-forward pod/program-8 3000:3000", which shows "Forwarding from 127.0.0.1:3000 -> 3000" and "Handling connection for 3000".

```
1RV24MC087_SSDhanush@dhanush-Latitude: ~/program8  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program8$ kubectl apply -f nodejs-configmap.yaml  
configmap/nodejs-app-config created  
pod/program-8 created  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program8$ kubectl get pods  
NAME      READY   STATUS    RESTARTS   AGE  
program-8 0/1     ContainerCreating 0          23s  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program8$ kubectl get pods  
NAME      READY   STATUS    RESTARTS   AGE  
program-8 0/1     ContainerCreating 0          25s  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program8$ kubectl get pods  
NAME      READY   STATUS    RESTARTS   AGE  
program-8 0/1     ContainerCreating 0          28s  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program8$ kubectl get pods  
NAME      READY   STATUS    RESTARTS   AGE  
program-8 0/1     ContainerCreating 0          31s  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program8$ kubectl get pods  
NAME      READY   STATUS    RESTARTS   AGE  
program-8 0/1     ContainerCreating 0          38s  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program8$ kubectl get pods  
NAME      READY   STATUS    RESTARTS   AGE  
program-8 0/1     ContainerCreating 0          54s  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program8$ kubectl get pods  
NAME      READY   STATUS    RESTARTS   AGE  
program-8 1/1     Running    0          85s  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program8$ kubectl port-forward pod/program-8 3000:3000  
Forwarding from 127.0.0.1:3000 -> 3000  
Forwarding from [::1]:3000 -> 3000  
Handling connection for 3000  
Handling connection for 3000
```



A web browser window showing the address bar with "localhost:3000". The page content displays "Hello program8".

```
localhost:3000  
Hello program8
```